



Assessing Thailand's Exposure to Taiwan Semiconductor Supply Chain Disruption

Independent Research

Thailand's manufacturing boom created an unmeasured semiconductor supply chain vulnerability

Rising import dependency: HS8542 imports from the Taiwan proxy rose from \$ 2.21B to \$ 10.65B (2014–2024). Share of total imports jumped from 22.8% to 43.6%.

Geopolitical risk: Cross-strait tensions and military exercises in 2022–2024 have elevated Taiwan semiconductor disruption risk for global supply chains.

2021 chip shortage precedent: Partial semiconductor shortages halted automotive plants, delayed electronics, and disrupted HDD production globally — a documented template for downstream impact.

Measurement gap: No public analysis had quantified Thailand's industry-level semiconductor exposure using trade data. This project addresses that gap.

Research Question

How exposed are Thai industries to Taiwan-related semiconductor supply chain disruption, and which face the highest impact?

4 Industries Analyzed

Electronics \$ 52.9B • chip share 25%

Automotive \$ 53.0B • chip share 4%

HDD / Storage \$ 13.4B • chip share 15%

Home Appliances \$ 29.9B • chip share 6%

A three-component framework converts UN Comtrade trade data into industry-level disruption exposure estimates

DATA SOURCE

UN Comtrade
HS8542 Integrated
Circuits

2014–2024
47 partners
1,178 records

Tools:
Python / pandas
Jupyter Notebook

TPDI

**Taiwan Proxy
Dependency Index**

Formula

$$\frac{\text{Imports (Taiwan proxy)}}{\text{Total HS8542 Imports}} \times 100$$

Output

Trend 2014→2024
22.8% → 43.6%
Phase breakdown

RAR Model

**Industry
Revenue-at-Risk**

Formula

$$\begin{aligned} &\text{Production Value} \\ &\times \text{Chip Share (base)} \\ &\times \text{Taiwan Dependency} \end{aligned}$$

Output

Total \$ 8.35B
across 4 industries
(low / base / high)

Scenarios

**Disruption
Impact Analysis**

Formula

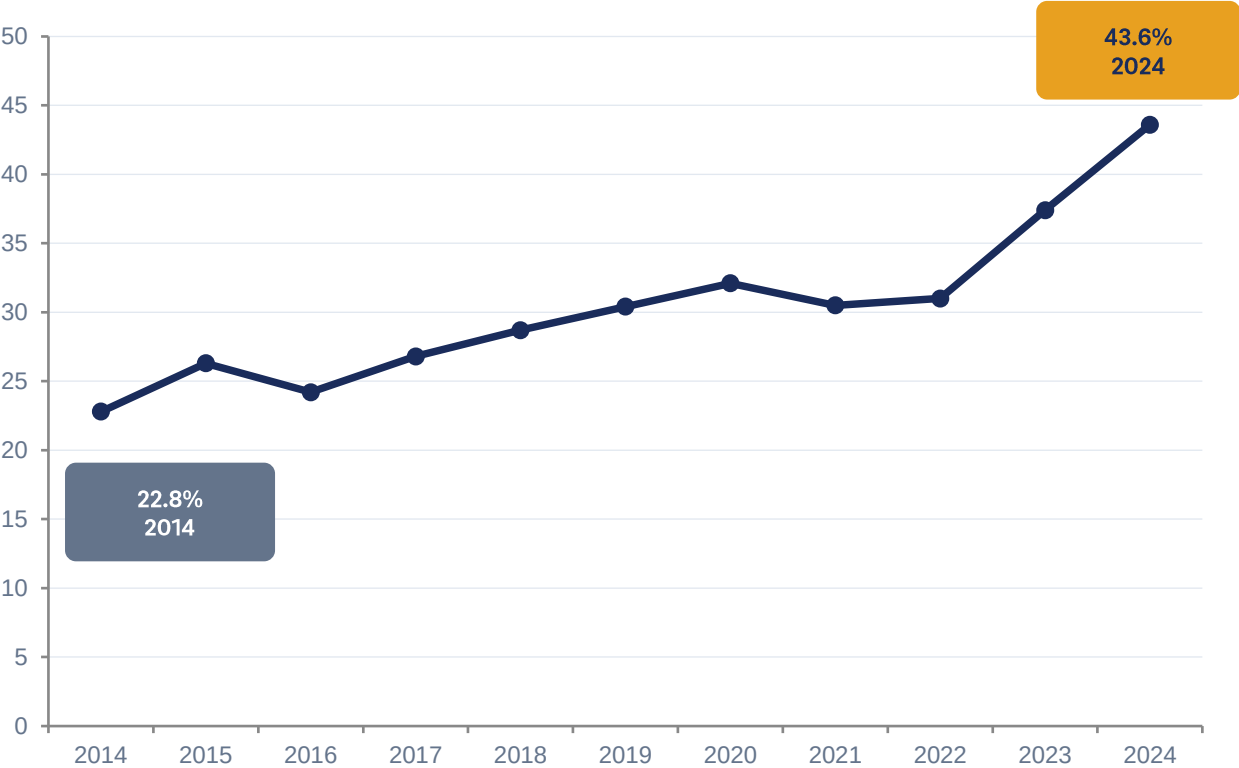
$$\text{Base RAR} \times \text{Severity}$$

10% / 20% / 40%
(historical anchors)

Output

Mild \$ 0.83B
Mod \$ 1.67B
Severe \$ 3.34B

Taiwan-proxy dependency nearly doubled from 22.8% to 43.6% over 10 years, accelerating sharply since 2022



Phase 1 • 2014–2020

Gradual rise
+9.3% over 6 years

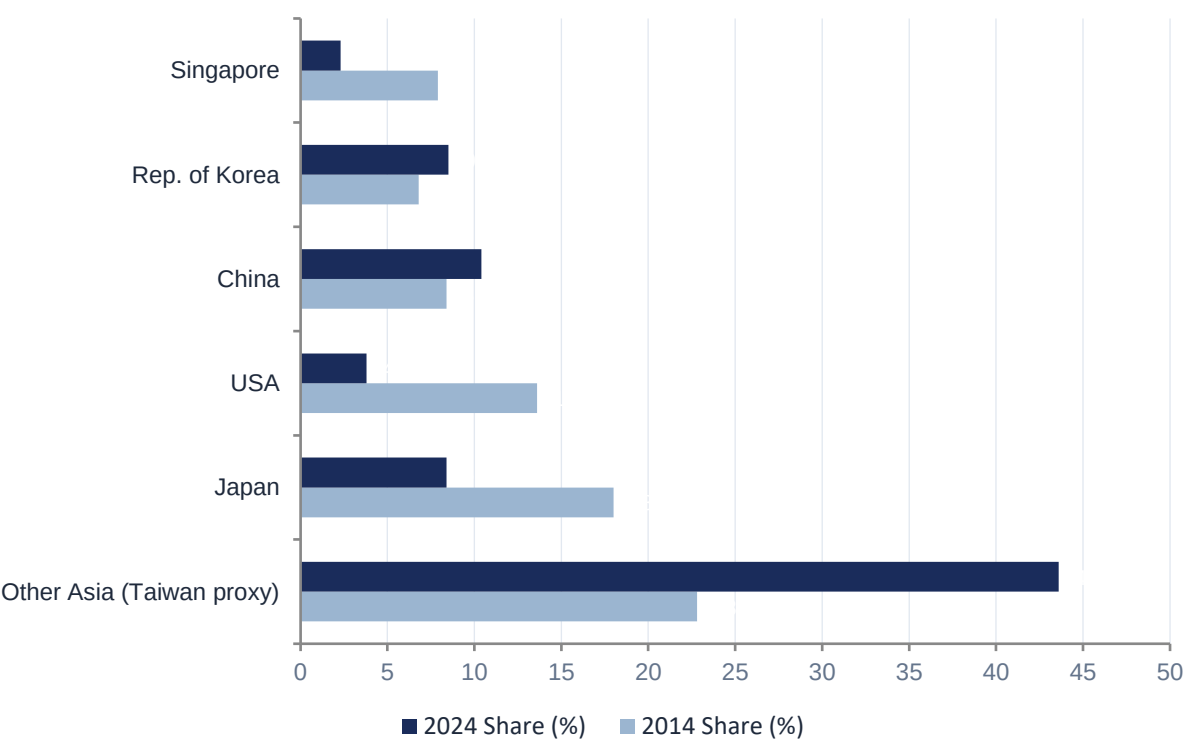
Phase 2 • 2021

Brief moderation: surge in total imports diluted Taiwan share

Phase 3 • 2022–2024

Acceleration: +12.6% in 2 years
Taiwan imports grew 2× total rate

As Taiwan-proxy share rose to 43.6%, Japan fell from 18.0% to 8.4% and USA from 13.6% to 3.8%



Other Asia, nes (Taiwan proxy)

+20.8%

22.8% → 43.6%

Dominant & growing

Japan

-9.6%

18.0% → 8.4%

Key alternative declined

USA

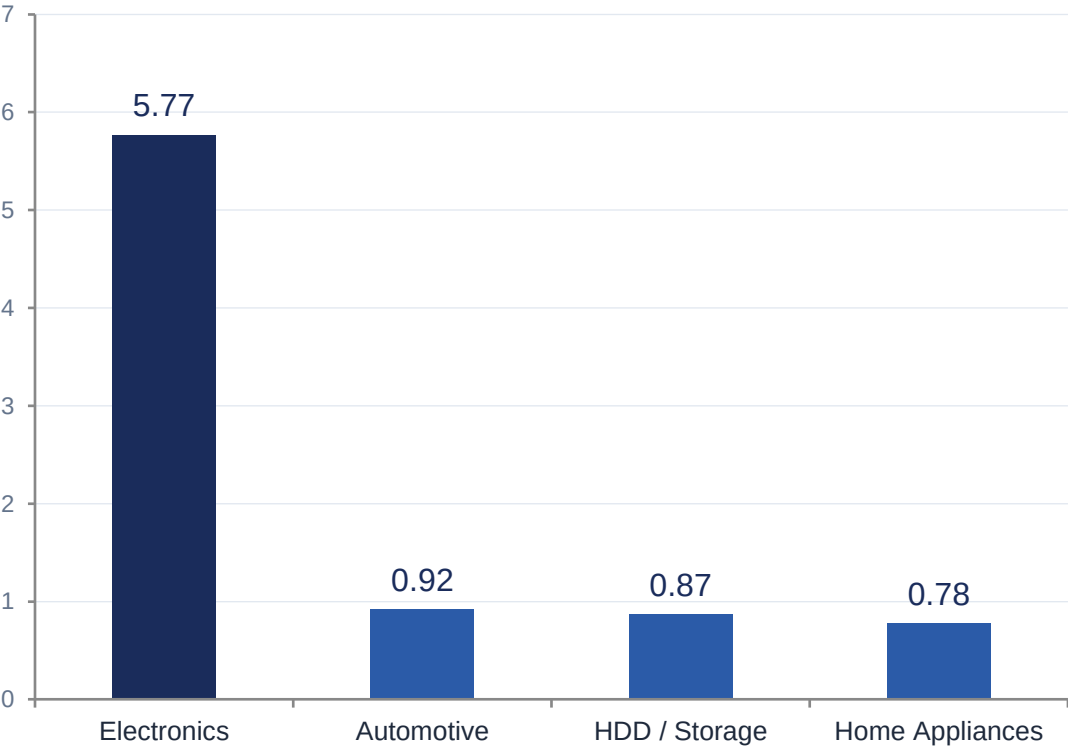
-9.8%

13.6% → 3.8%

Steepest relative drop

Thailand has fewer alternatives today
short-term substitution of Taiwan-proxy
chips is harder than the raw TPDI implies.

Electronics dominates: USD 5.77B base-case Revenue-at-Risk — approximately 69% of USD 8.35B total exposure



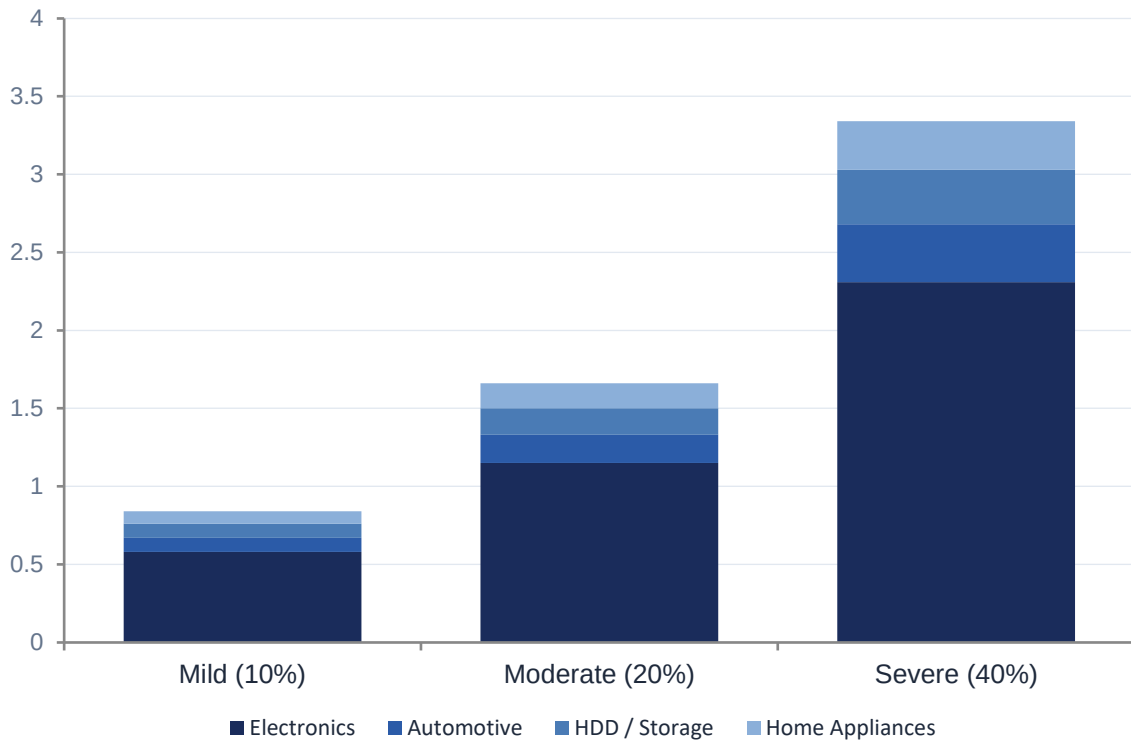
\$ 8.35B

Total base-case Revenue at Risk
across 4 industries

Industry Breakdown

Electronics	\$5.77B	~69%
Automotive	\$0.92B	~11%
HDD / Storage	\$0.87B	~10%
Home Appliances	\$0.78B	~9%

Disruption impact ranges from \$ 0.83B (mild) to \$ 3.34B (severe) — anchored to 2021 chip shortage data



Mild • 10% severity

\$ 0.83B

Ford CFO guidance: 10–20%
Q1 production loss (2021)

Moderate • 20% severity

\$ 1.67B

Upper end of Ford range;
broad industry disruption
(2021 chip shortage)

Severe • 40% severity

4 3.34B

Toyota: 40% global
production cut,
September 2021

Three distinct vulnerability profiles require tailored risk strategies — not a one-size-fits-all approach

High Absolute Exposure

Electronics

\$ 5.77B | ~69% of total

Root cause:

Large production base (\$ 52.9B) combined with high chip intensity (25% cost share) = dominant exposure

Strategy:

Prioritize for national resilience strategy. Highest aggregate risk reduction per dollar of mitigation effort.

Strategic Bottleneck Risk

Automotive + HDD / Storage

\$ 1.79B combined RAR

Root cause:

Low cost share (4–15%) understates risk: one missing chip halts an entire high-value production line

Strategy:

Apply a bottleneck-risk lens: classify chips by production impact, lead-time uncertainty, and substitutability — not unit cost.

Moderate but Rising

Home Appliances

\$ 0.78B RAR | rising intensity

Root cause:

Lowest absolute exposure today but smart-appliance chip content is rising with inverter and IoT adoption

Strategy:

Monitor chip intensity trend annually. Re-evaluate cost-share assumption as connected-appliance share grows.

Five actions reduce Thailand's semiconductor supply chain vulnerability – from immediate to long-term

1

Strategic Inventory Buffering

Short-term

Target: Auto, HDD, Electronics

Classify chips by bottleneck criticality. Build safety stock based on production impact, lead time, and substitutability – not unit cost.

2

Industry Risk Monitoring Dashboard

Short – Medium

Target: All 4 industries

Track TPDI and supplier concentration annually with UN Comtrade data. The Python framework from this project is ready to operationalize.

3

Supplier Diversification

Medium-term

Target: Electronics, Auto, HDD

Qualify secondary suppliers in Korea, Japan, and Malaysia for mature-node chips. Establish dual-sourcing for high-criticality components.

4

Annual Scenario Planning

Recurring (annual)

Target: All 4 industries

Institutionalize mild/moderate/severe disruption exercises. Link severity levels to procurement triggers and production contingency plans.

5

Semiconductor-Adjacent Capability Dev.

Long-term

Target: Electronics, Automotive

Attract OSAT, advanced PCB, and power electronics via BOI incentives. Build semiconductor engineering workforce through universities.

These findings extend beyond Thailand the framework is replicable for ASEAN peers and future research



ASEAN-wide relevance

Vietnam, Malaysia, and the Philippines likely face similar Taiwan-proxy dependency patterns as they expand electronics and automotive manufacturing. Coordinated ASEAN-level supply chain resilience could achieve greater risk reduction than individual country actions.



Window for action is narrowing

At 43.6% TPDI and rising, technical substitutability of Taiwan-related supply may decline as manufacturing becomes more chip-intensive. Early diversification is less costly than reactive responses after disruption.



Open-source replicable framework

This project provides a reproducible methodology that uses only public data: UN Comtrade analysis, a RAR model, and scenario analysis. The full Python codebase, cleaned datasets, and Jupyter notebooks are open source on GitHub – adaptable to HS8541, other ASEAN countries, or other supply chain risk topics.

Thailand's semiconductor exposure is real, growing, and measurable — the window for action is now

01 Dependency nearly doubled

Taiwan-proxy share rose from 22.8% to 43.6% (2014–2024), with acceleration after 2022.

02 Alternatives have declined

Japan fell 18.0%→8.4%, USA 13.6%→3.8% — Thailand has fewer short-term substitution options than a decade ago.

03 Electronics leads; bottlenecks matter

\$ 5.77B base RAR (~69%) from electronics; automotive and HDD carry hidden bottleneck risk beyond cost-share figures.

04 Disruption impact is economically significant

\$ 0.83B (mild) to \$ 3.34B (severe) — exposure estimates, not forecasts, but too large to ignore.

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Industrial Engineering • Chulalongkorn University • June 2026
Python • pandas • Plotly • UN Comtrade / WITS • Jupyter Notebook

[GitHub Repository](https://github.com/methakanb/taiwan-semiconductor-risk-thailand)

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